# OSPF Redundant Enterprise Network Design

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## 1 Introduction

This project demonstrates the implementation of a fault-tolerant enterprise network using the Open Shortest Path First (OSPF) dynamic routing protocol in Cisco Packet Tracer. The network topology was designed with multiple routers, redundant paths, web servers, and a DNS server to simulate a real-world enterprise network environment.

The main objective of this project was to provide reliable communication between networks and maintain connectivity during router failures using dynamic routing techniques.

## 2 Project Objectives

- Design a scalable enterprise network topology
- Implement OSPF dynamic routing between routers
- Configure redundant routing paths for fault tolerance
- Enable communication between client PCs and servers
- Simulate router failure and verify automatic rerouting
- Validate network connectivity using routing and traceroute commands

## 3 Network Topology

The network topology consists of the following components:

- 5 Cisco Routers
- 2 Cisco Switches

- Multiple PCs
- 2 Web Servers
- 1 DNS Server
- Redundant WAN Links

The routers are interconnected using multiple paths to provide redundancy and improve network reliability. OSPF dynamic routing was configured across all routers to allow automatic route learning and path selection.

# 4 IP Addressing Scheme

## 4.1 Client LAN

| **Device** | **IP Address** |
|---|---|
| PC0 | 192.168.10.2/24 |
| PC1 | 192.168.10.3/24 |
| Default Gateway | 192.168.10.1 |

## 4.2 Server Network

| **Device** | **IP Address** |
|---|---|
| Router Gateway | 192.168.11.1/24 |
| Web Server 1 | 192.168.11.2/24 |
| Web Server 2 | 192.168.11.3/24 |
| DNS Server | 192.168.11.4/24 |

# 5 OSPF Configuration

OSPF was configured on all routers to dynamically exchange routing information and maintain efficient communication across the network.

## 5.1 Features Implemented

- Dynamic route learning
- Automatic path recalculation
- Router neighbor discovery

- Redundant routing paths
- Network convergence during failures
- End-to-end connectivity between networks

# 6 Fault Tolerance and Redundancy

To test the reliability of the network, one router was intentionally powered off to simulate a real network failure.

After the router failure:

- OSPF automatically recalculated routes
- Traffic was redirected through an alternative path
- PCs successfully maintained communication with servers
- No manual configuration changes were required

Traceroute tests confirmed that OSPF dynamically selected a new path after the failure occurred.

# 7 Verification and Testing

The following commands were used to verify the network configuration and routing behavior:

```
show ip route
show ip ospf neighbor
show ip protocols
ping
tracert
```

The test results confirmed successful routing table exchange, neighbor establishment, and continuous connectivity during network failures.

# 8 Technologies Used

- Cisco Packet Tracer
- Cisco Routers

- Cisco Switches
- OSPF Routing Protocol
- DNS Services
- Web Servers
- TCP/IP Networking

# 9 Key Learning Outcomes

This project provided practical experience in:

- OSPF Dynamic Routing
- Enterprise Network Design
- Fault-Tolerant Network Architecture
- Dynamic Route Convergence
- Network Troubleshooting
- Routing Table Analysis
- Redundant Path Configuration
- Server Connectivity Testing

# 10 Future Improvements

Possible future improvements include:

- VLAN segmentation
- DHCP integration
- NAT configuration
- Access Control Lists (ACLs)
- SSH remote management
- IPv6 support
- Multi-area OSPF implementation

# 11 Conclusion

This project successfully demonstrated the implementation of a fault-tolerant enterprise network using OSPF dynamic routing. The use of redundant paths and automatic route recalculation ensured continuous communication even during router failures.

The project provided valuable practical experience in enterprise routing, network redundancy, and dynamic routing protocols, which are essential skills for modern network engineering environments.